admits of the subsidence of any irritative inflammation about the seat of the stricture so that exudates are absorbed and muscular spasm is lessened. It is therefore impossible not to conclude that though the operation of gastrostomy successfully executed is only palliative, it is nevertheless likely to be the means of prolonging life longer if performed early than when the period of necessity has been reached; in other words, before and not after difficulty in swallowing fluids has occurred.

Glasgow.

VACCINATION RASHES.

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IT is a question of some importance what precisely may be the influence of vaccination in the production of various skin eruptions. Has vaccinia such an effect upon the human organism as to render the skin more liable to become affected by rashes, or are we to regard as a mere coincidence the fact, which is undoubted, that children, and indeed adults, may after vaccination suffer from various forms of skin eruptions? The truth would appear to be between these two extremes. Vaccination is undoubtedly responsible directly for some forms of eruption, and indirectly in a remote degree for others; but it is manifestly unfair to ascribe to vaccination an eruption which appears weeks, or even months after the lesion produced by the operation has soundly healed. In my experience post-vaccinal rashes are uncommon—that is to say, such rashes as may occur within a period of time in which it may reasonably be supposed that the course of vaccinia has in some way or other been instrumental in causing their appearance. What this limit of time after vaccination may be at which eruptions of the skin may appear, due indirectly or directly to the operation, it is difficult to state precisely. Dr. Crocker states that in no case can an eczema be due to vaccination where the vaccine pustule has completely healed before the appearance of the eczema. Personally I should hesitate to ascribe any rash as due to vaccination except vaccino-syphilis, when after examination the site of the vaccination is found soundly healed, which may roughly be taken as occurring from 4 to 5 weeks after the operation. The results of recent researches have made it clear exactly what vaccination is and we are consequently better able to gauge its likelihood of causing skin affections as compared with other acquired diseases. That true vaccinia may result from the inoculation of a modified attenuated variolous lymph cannot now be questioned. Variolous lymph by its passage through a series of calves becomes so attenuated and modified that it can no longer produce violent constitutional effects with a generalised eruption, but only a slight constitutional disturbance and an eruption limited to the points of inoculation, and a condition of the organism is produced which renders it immune against variola or vaccinia for a variable period of time. How this immunity is produced, whether by the organisms of the vaccine virus multiplying in the body and by their life and growth producing complex chemical bodies, which by absorption produce a change in the human tissues, rendering them immune, or, as Dr. Cory suggests, whether the whole production of such chemical bodies takes place in the vaccine vesicles, may be an interesting speculation, but leads nowhere. What interests us more is the certain fact that for our purpose vaccinia may be practically classed amongst the other exanthemata, that it is a constitutional disease which more or less profoundly affects the whole of the tissues of the human organism in which it occurs. That some tissues in the same organism are affected more profoundly than others is a well-known clinical feature in every constitutional disease and it would appear quite easy to conceive that in certain individuals vaccinia may produce a condition of cell change and tissue alteration in the skin causing the appearance of a rash, the type of such rash varying with the individual peculiarity of the skin—a peculiarity not of texture, complexion, &c., but physiological and vital. These remarks only refer to what are termed "pure vaccination rashes," that is, due to the inoculation of pure vaccine virus and not to extraneously introduced germs. The connexion between vaccination and the eruptions due to extraneously introduced germs is absolutely one of cause and effect and requires here no further discussion. Owing to the enormous percentage of children vaccinated in the present day (practically in London over 90 per cent.) had vaccinia much influence as a causative agent in the production of skin affections surely they would have increased *pari passu* with vaccination. This I venture to doubt, for a reference to writers of the pre-vaccination days certainly leaves one with the impression that such a disease as infantile eczema was as common then as now. As far as I have been able to ascertain there are no published figures as to the possible frequency with which post-vaccinal rashes may occur and, in fact, the whole literature of the subject is very meagre. As a possible result of the paucity of such literature and references in standard works many rashes are ascribed to vaccination, not only by the parents of children, but by medical practitioners, which have absolutely no connexion with vaccination either near or remote. A case of generalised eczema wrongly ascribed by a medical practitioner to vaccination is a centre from which all kinds of anti-vaccinationist poison emanate. The possible dangers of vaccination are absolute and quite real enough without the invention of bogies to frighten weak-minded people about an operation, which is probably the most important to the community at large of any we are ever called upon to perform.

It is true that "from their rarity post-vaccinal rashes are unimportant," to quote the words with which one writer dismisses the subject, but it appears to be of the utmost importance that every medical practitioner should have a precise and definite knowledge of the subject so that he may successfully combat, on the one hand, the prejudices of the parents and the glaring misstatements of the anti-vaccinationists on the other, and to be able readily to demonstrate the fallacy of the *post hoc propter hoc* argument so often and so dangerously applied to vaccination and subsequent rashes.

Vaccination rashes have been variously classified. Dr. Crocker's modification of the one by Mr. Malcolm Morris would appear to be the most complete, and, with an addition of my own, is the one I shall use as a basis for my remarks.

CLASSIFICATION OF VACCINATION RASHES.

Group I.—Eruptions resulting from pure vaccine inoculation. *A*, Secondary local inoculation of vaccine. *B*, Eruptions before the vesicles form, urticaria, erythema multiforme, vesicular and bullous eruptions. *C*, Eruptions after formation of the vesicles due to the absorption of the virus: (*a*) morbilliform, scarlatiniform and diffuse erythema, erythema multiforme, vaccine lichen, purpura; and (*b*) generalised vaccinia. *D*, Sequelæ of vaccination, eczema, psoriasis, pemphigus, urticaria, and congenital syphilitic rashes.

Group II.—Eruptions due to the vaccine plus some other virus. *A*, Introduced at the time of vaccination: (*a*) producing local disease, impetigo contagiosa, or other form of superficial inflammation; and (*b*) producing constitutional disease, syphilis, leprosy, and tuberculosis. *B*, Introduced after development and rupture of the vesicles (about the eighth day), erysipelas, cellulitis, impetigo contagiosa, furunculosis, gangrene local or disseminate, pyæmia.

This classification shows how varied in character post-vaccinal rashes may be and that they are almost as numerous as there are rashes to which an individual is liable. What determines the precise nature of the skin manifestation is, as I suggested above, a physiological and vital peculiarity of the tissue, or what is more popularly called "the idiosyncracy of the patient." That this must be is evident from the fact that of several children vaccinated from one source only one may exhibit post-vaccinal rash, or if there be two the rashes are of a different nature. The fact also that post-vaccinal rashes are even more frequent (according to the authorities) after vaccination with calf lymph than with humanised lymph points in a similar direction. It is a well-known clinical fact that local irritations in susceptible individuals may produce generalised skin eruptions. The wet napkin in infants, a chronic nasal catarrh, profuse perspiration, the use of certain irritants for the purposes of trade, and many others may, and indeed do, produce an inflammatory reaction in the skin by no means confined to the point of contact. Equally, then, vaccination in certain cases, the local irritation in the arm having superadded to it the constitutional effects of vaccination, is liable to produce skin eruptions the exact type of which depends upon some peculiarity of the individual tissues, about which we know nothing and can surmise less We may now discuss more or less in detail the classification of post vaccinal rashes given above.

In Group I. under section *A* we have the secondary inoculation of vaccine. This may be auto-inoculation or accidental as apart from vaccination. As an instance of the latter some time ago I saw a female, seventy-two years of age, with a typical vaccine vesicle on the cheek. This was the result of nursing an infant whose vaccine vesicles had been opened on the eighth day, whilst on her cheek was a scratched papule which allowed the introduction of the lymph from the child's arm. Many such accidents have been recorded: vaccine vesicles on the eyelids, where their resemblance to primary syphilitic sores renders them important, on the tongue, and a remarkable case where a child suffering from eczema of the face and scalp sleeping with an elder child who had recently been vaccinated was inoculated by the latter, so that the whole eczematous area became the seat of vaccinal vesicles. It is a noteworthy point in secondary auto-inoculation that the secondary vesicles are more rapid in their formation than the primary and reach maturity almost as quickly.

Subsection B.—Urticaria may follow vaccination in infants just as in adults it may arise from apparently most trivial causes. It, however, differs from the urticaria of adults and presents the well-known clinical picture of urticaria papulosa, an affection most difficult to treat, resisting more or less effectually all therapeutic measures to yield at length to the lapse of time. Of the erythema multiforme, vesicular and bullous eruptions mentioned in the classification, but few writers even mention them, and then only by name; personally I have never seen such post-vaccinal rashes occurring before the formation of the vesicles.

Subsection C.—In this subsection the eruptions are due to the absorption of the vaccine virus and divide themselves naturally into two groups: (*a*) morbilliform, scarlatiniform, and diffuse erythema, erythema multiforme, vaccine lichen, purpura, and (*b*) generalised vaccinia. Under (*a*) it is curious to note that the morbilliform, scarlatiniform, diffuse erythema and purpura may all occur preceding the appearance of the exanthematic manifestation of small-pox. May not a peculiar condition of tissue in the organism reacting to the poison of variola react similarly to vaccinia? Certain it is that in both cases they occur but rarely. With the title roseola, erythema vaccinal (French), roseola vaccinia (German), occurs a fairly common erythematous eruption which some observers describe as the commonest form of post-vaccinal eruption. It appears about from the eighth to the eleventh day, but according to Hebra as early as the third or as late as the eighteenth day after vaccination. Kaposi describes it as the most frequent complication of vaccinia and states that it commences around the vaccine vesicle and gradually spreads over the body. In my experience it is a rosy red rash not unlike measles, the macules varying from a pin's point in size to the size of a five-shilling piece. It is not attended with any constitutional symptoms and fades after three or four days without any desquamation. Dr. Radcliffe Crocker describes a vaccine lichen which he considers the commonest post-vaccinal eruption. According to his description it may be papular, papulo-vesicular, pustular, rarely bullous, appearing on from the fourth to the eighteenth day, most usually on the eighth day. It commences on the arms and by successive crops may spread over a considerable part of the whole body, evenly distributed, sometimes becoming circinate. The papules are acuminate, pin's-point sized, bright red, and may remain so to the end, usually discrete, but occasionally coalesce into patches, often many of the papules are crowned with small vesicles or pustules and occasionally the whole eruption may be vesicular, such vesicles coalescing and becoming bullous, as recorded by Behrend. The rash lasts usually a few days, but occasionally successive crops appear attended by considerable itching, resembling Hutchinson's varicella prurigo. This account, taken from Dr. Radcliffe Crocker's book on Diseases of the Skin, corresponds closely with the vaccinal miliaria described by Dauchez in his work on vaccination rashes. Dr. Colcott Fox suggests that the different phases of vaccine lichen probably represent stages of the same inflammatory process. Both Behrend and Crocker have recorded cases of typical post-vaccinal erythema multiforme.

A purpuric rash may occur after vaccination in cachectic children. Cases have been reported by Gregory and Dauchez but are extremely rare. Under (*b*) generalised vaccinia is probably the most important of Subsection C. Under exceptional circumstances, chiefly after vaccination with calf lymph, the vaccine eruption, instead of being confined to the points of inoculation, is more or less widely spread over the body. It is still a debatable question whether these cases are due to general blood infection or whether the secondary vesicles are the result of auto-inoculation. According to the various writers the rash appears about the eighth day and its course resembles that of a vaccine vesicle. In the fatal cases inflammation and sloughing take place at the site of the vesicles, producing a condition exactly resembling varicella gangrenosa. In this country Mr. Hutchinson has seen and recorded several such cases. Dr. Cory, the Local Government Board Inspector, in his recent work describes cases similar to varicella gangrenosa as due to disseminated tubercle, but does not mention generalised vaccinia. Dr. Colcott Fox states that they are always the result of auto-inoculation. French dermatologists are generally agreed that there are two distinct kinds of generalised vaccinia (*a*) appearing spontaneously and (*b*) the result of auto-inoculation.

Subsection D comprises the sequelæ of vaccination—viz., eczema, psoriasis, pemphigus, urticaria, and congenital syphilitic rashes. Of these from its frequency eczema is important. Dermatologists differ in their opinions as to the frequency of eczema caused by vaccination, but all appear to agree with Dr. Crocker that no case should be ascribed to vaccination which has appeared after the vaccination has soundly healed. This consensus of opinion clears the ground and effectually disposes of the very large majority of infantile eczemas ascribed by parents to vaccination. A curious well-authenticated clinical observation is that vaccination sometimes cures an attack of eczema, and indeed has been recommended as a legitimate form of treatment not only in adults but in children. In the latter, however, when the eczema is acute it is attended with great risk owing to the possibility of auto-inoculation. There can be no doubt that in predisposed people eczema may follow vaccination, as it might any other form of dermatitis, however produced, but there is nothing in vaccinia *per se* which can cause an outbreak of the disease. What this predisposition to eczematous forms of dermatitis consists of, in the present state of our knowledge, it is impossible to say definitely. It has been explained as "a want of resistance in the epithelial structure" or as "vulnerable epithelium," but such explanations do not assist us much. Certain it is that in the predisposed any slight local irritation may give rise to a widespread attack of eczema. When following vaccination eczema may commence in the region of the vesicles and gradually spread, or, more rarely, a large area of the skin may become suddenly attacked. It usually appears on or about the tenth day of the vaccinal period and in its course and amenability to treatment would appear exactly to resemble eczema of other origin.

Post-vaccinal psoriasis is sufficiently rare to be a clinical curiosity. Chambard has described one case and Rohè two acute cases. Pemphigus is barely mentioned in French and English literature as following vaccination. The German dermatologists describe it, however, as occurring occasionally in weakly infants and they differentiate it from ordinary pemphigus due to cachexia on account of the fact that the prognosis in post-vaccinal pemphigus is distinctly favourable, physically their appearance being identical. Urticaria may occur as a sequela in point of time, but its association with vaccination would appear to be one of coincidence.

Congenital syphilitic rashes I place under the head of sequelæ, although apparently none of the various classifications published include them. In view of their extreme importance, both from a medical and lay point of view, their possible occurrence should be always borne in mind. That definite syphilitic rashes occur after vaccination it is impossible to deny, and it is of the utmost importance that not only should they be recognised as such, but that they should be attributed to their real source—namely, congenital infection. In Group II. syphilis occurring as the result of vaccination due to vaccine lymph plus the syphilitic virus will be treated, but in this group the clinical facts are quite dissimilar. The usual history obtained is that on or about the tenth day after vaccination an obviously syphilitic rash appeared. It hardly requires, except for the fact that so many misstatements have been made, that it should be pointed out that such a syphilitic rash is due, not to syphilis introduced at the time of vaccination, but to a congenital infection. No rash in the case of inoculated syphilis ever appears before, at the very earliest, fifty days after inoculation. Apparently in the case of a child suffering from congenital syphilis, vaccinia produces enough local irritation and general constitutional disturbance to cause the appearance of skin manifestations, vaccination simply determining the date of the appearance of the skin lesions. Notwithstanding the fact that calves are absolutely immune against syphilis, congenital syphilitic rashes much more frequently appear after vaccination with calf than with humanised lymph.

Group II. consists of post-vaccinal rashes more or less preventable. Certainly Section A, those due to a virus plus the vaccine virus introduced at the operation of vaccination, are, with the exercise of ordinary precautions, absolutely preventable, and when occurring indicate culpable negligence on the part of the operator.

Subsection (a).—As a result of the introduction of pus cocci impetigo contagiosa may occur. This is a very rare accident, accidental inoculation taking place much more frequently after rupture of the vaccine vesicles. Short of true impetigo contagiosa slight forms of superficial inflammation may occur as a result of the introduction of one or other of the organisms found in pus.

Subsection (b).—The occasional inoculation of syphilis with vaccinia cannot be denied and is the great weapon in the hands of the anti-vaccinationists, but it has happened so rarely as practically to become a clinical curiosity, and with the introduction into general use of calf lymph will disappear altogether. When syphilis is inoculated simultaneously with vaccine virus, according to Fournier three alternatives present themselves: (1) syphilis is not transmitted but vaccinia is; (2) syphilis is transmitted and vaccinia fails; and (3) both are transmitted, some points of inoculation giving rise to typical vaccine vesicles, whilst later a syphilitic chancre or chancres appear at a point of inoculation when vaccinia has failed. Hutchinson states that a syphilitic chancre may follow typical vaccine vesicles in the same site. Fournier gives an excellent table in his clinical lecture on the subject, as to the differential diagnosis of vaccino-syphilis which is too long to give here. Briefly, he points out that the syphilitic chancre is limited to one or two points of inoculation, inflammation is slight, loss of substance is superficial, and the parchment induration is typical, not inflammatory. The acquired syphilitic rash appears at the earliest from fifty to ninety days after vaccination and requires in every case the existence of a specific sore at the site of the vaccination. In fact, vaccino-syphilis is in no way different from syphilis acquired in other ways and follows precisely the same course.

In this country the question of the introduction of leprosy with vaccinia is purely academic and the reported evidence from other countries is very conflicting. Tuberculosis is a disease which is confidently asserted by the anti-vaccinationists as likely to be introduced by vaccination. In Germany many experiments have been made with lymph from tuberculous patients with uniformly negative results. Attempts to inoculate the skin of guinea-pigs—animals most susceptible to tuberculosis—have invariably failed. That the skin of the human organism, however, is susceptible to the inoculation of tubercle cannot be denied, as cases have been reported where a local tuberculosis has been produced from tattooing and the like. Recently at the Dermatological Society of London a case was shown in which, although there was no microscopical or bacteriological evidence, yet the majority of those present agreed that the appearance was that of a local tuberculosis the result of vaccination. The question, however, of the transmission of the three diseases, syphilis, leprosy, and tuberculosis, by vaccination is probably solved by the introduction of glycerinated calf lymph by the new Vaccination Act of the present Government and ought to go a long way towards disarming what might be a legitimate criticism of vaccination.

Section B of Group II. comprises rashes due to the introduction of a virus after the formation and rupture of the vaccine vesicles. Of these by far the most important is erysipelas. It is possible that the pathogenic organism of erysipelas may be introduced into the patient on the eighth day by the operator opening the vesicles, but the accident must be excessively uncommon. After the opening or rupture of the vesicles the introduction of the erysipelas organism is not so uncommon and in the large majority of cases occurs when the child is living in bad hygienic surroundings. The characteristic rash of erysipelas generally appears within 24 hours after inoculation around the vesicles and spreads rapidly. In young infants it is very fatal and it was on this account that in the new Vaccination Act it was first proposed to raise the age limit at which vaccination must have been performed from three months (as now) to one year, statistics showing that the death-rate from erysipelas gradually diminishes with each succeeding month of the first year of life. Cellulitis mentioned in the classification is probably erysipelas complicated with pus cocci. Impetigo contagiosa is not uncommon after the rupture of the vesicles, around which it spreads and by the child's fingers may be carried to other parts of the body. Furunculosis is described as occurring as the result of the absorption of pus cocci into the system, their dissemination and ultimate arrest in the skin.

Gangrene local or disseminate. A local gangrenous action the result of septic poisoning may sometimes be seen at the site of the vaccination. A large black slough forms, which when detached leaves a deep ulcer which takes weeks to heal. Clement Lucas has reported a fatal case in an infant three months old. Disseminated gangrene the result of vaccinia is an exactly similar condition to that described as varicella gangrenosa. The lesions commence as a pin's-head sized papulo-pustule generally on the vaccinated arm and appear all over the body in crops. The pustule ruptures, a scab forms with an inflammatory aerola, and the centre becomes a black slough. Barlow has pointed out that the majority of cases, whether occurring after varicella or vaccinia, are associated with tuberculosis.

Cases of pyæmia after vaccination have been reported. As clinical curiosities may be mentioned the case published by Diday where a growth of hair took place around the cicatrix of a vaccination performed on the anterior part of the thigh of an infant eleven months old, and the occasional occurrence of keloid in the vaccination cicatrix, two cases of which have been reported.

Acacia-road, N.W.

NOTES ON THE

LOCALISATION OF A CENTRE IN THE BRAIN CORTEX FOR RAISING THE UPPER EYELID.

BY W. HARTLEY BUNTING, M.D., F.R.C.S. EDIN.

ALTHOUGH most of the motor centres in the brain cortex have for some time been accurately defined the centre for raising the upper eyelid is as yet uncertain. The very existence of such a centre has indeed been doubted by some, who hold that the centre for the levator palpebræ superioris along with that for all the extrinsic muscles of the eyeball is intra-cortical and is situated somewhere in the region of the corpora quadrigemina. I think, however, that there is strong evidence to show that a cortical centre does exist and that it is situated far forward in the general motor area. There are some difficulties which present themselves in the way of experimentally localising this centre. In the first place, since the eyelids on the two sides under normal conditions act together, one finds that the centres in the two halves of the brain for raising the lid are so intimately associated that compensation after destruction of the centre on one side readily takes place and ptosis is frequently not observed at all, or, if noted, it is quite temporary, the opposite side taking on the function; this theory was first advanced by Wernicke. Horsley as the result of the study of a clinical case arrived at the same conclusion, and he says that when cortical ptosis exists from damage to one side only of the brain cortex the centres for the elevators of the upper lid are not so closely associated as usual. Again, fallacies have doubtless arisen from the fact that all the external ocular muscles, the levator palpebræ included, may be excited to coördinated movements by vigorous stimulation in many parts of the brain cortex; moreover, it is but rarely that one has the opportunity of making a necropsy in a case of cortical ptosis, or if such an opportunity does occur, the lesion is probably so extensive as to render the results in this direction negative.

Landouzy and Grasset advanced the theory that a centre for movement of the upper lid could be localised in the angular gyrus. The lesions, however, in Landouzy's case